

Introductory Material

In this experiment you will measure the magnitude of a magnetic dipole moment $\vec{\mu}$ in a magnetic field using three different methods. In Part I, you will balance the magnetic torque on a dipole in a magnetic field with the gravitational torque. In Part II, you will obtain the magnitude of the magnetic dipole moment by observing simple harmonic motion of a magnetic dipole in a magnetic field. In Part III, you will find the magnitude of a magnetic dipole moment by determining the force on the dipole in a non-uniform magnetic field.

In all three parts, the magnetic field will be created by passing an electric current through two coils of wire.

Magnetic Field

In order to calculate the magnetic field between the two current loops, some simplifying assumptions must be made in order to avoid some very complicated and tedious calculations. The major assumption will be to use an equivalent radius and equivalent separation between the coils to compensate for the fact that we have multiple coils of different radii and distances from each other. These values are given by the manufacturer of the apparatus as:

Equivalent radius, $R = 0.109$ m

Equivalent separation between the coils, $D = 0.138$ m

With these assumptions in hand, the Biot-Savart Law can be used to calculate the strength of the magnetic field a distance z away (perpendicular to the plane formed by the loop) from the center of a single loop of radius R with a current I :

$$B = \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}} \quad (9.1a)$$

If you have N loops, then

$$B = N \frac{\mu_0 I R^2}{2(R^2 + z^2)^{3/2}} \quad (9.1b)$$

Since the dipole will be located BETWEEN the two loops (see Figure 9-1), the distance to the point where we are calculating the field is only $D/2$. The manufacturer also states that each coil contains 195 loops (Don't forget that you have two coils, so $N=390$).

Magnetic Torque

You have learned that the torque produced on a dipole in a magnetic field can be expressed as:

$$\vec{\tau} = \vec{\mu} \times \vec{B} \quad (9.2)$$

and from mechanics we know that the torque produced by a weight on a rod in the earth's gravitational field is given by:

$$\vec{\tau} = \vec{r} \times m\vec{g} \quad (9.3)$$

If we can adjust these two so that the torques balance, the ball (see Figure 9-1) will not rotate. If the torques are indeed balanced, then the equations can be combined to give (using the fact that the cross product of two vectors can be written as the product of their magnitudes times the sine of the angle between them):

$$\mu B \sin \theta = rmg \sin \theta \quad (9.4)$$

since the angles involved in the cross products can be shown to produce the same sine, this simplifies to

$$\mu B = rmg \quad (9.5)$$

or

$$r = \left(\frac{\mu}{mg} \right) B \quad (9.6)$$

showing that the distance r the mass sits is directly dependent on the magnitude of the magnetic field.

However, since gravity exerts a torque on the rod as well as the weight, this equation should be modified to more accurately describe the combined weight-rod system. Specifically, we need to replace r (the location of the weight) with r_{cm} (the location of the center of mass of the rod-weight system) and m (the mass of the weight) with $m+m_{rod}$ (the total mass of the rod and weight combined). Thus we have the following equation:

$$r_{cm} = \left(\frac{\mu}{(m + m_{rod})g} \right) B$$

We can express the position of the center of mass in terms of the position of the weight and the position of the center of mass of the rod (r_{rod}) as follows:

$$r_{cm} = \left(\frac{m}{(m + m_{rod})} \right) r + \left(\frac{m_{rod}}{m + m_{rod}} \right) r_{rod}$$

Substituting this equation into our expression relating r_{cm} and B , we obtain the following relations:

$$\begin{aligned}
\left(\frac{m}{m+m_{rod}}\right)r + \left(\frac{m_{rod}}{m+m_{rod}}\right)r_{rod} &= \left(\frac{\mu}{(m+m_{rod})g}\right)B \\
\left(\frac{m}{m+m_{rod}}\right)r &= \left(\frac{\mu}{(m+m_{rod})g}\right)B - \left(\frac{m_{rod}}{m+m_{rod}}\right)r_{rod} \\
r &= \left(\frac{\mu}{mg}\right)B - \left(\frac{m_{rod}}{m}\right)r_{rod}
\end{aligned} \tag{9.7}$$

It is this last expression which we will use in the lab itself, as it removes the necessity of calculating the center of mass position of the rod.

Simple Harmonic Motion of a Magnetic Dipole in a Magnetic Field

A magnetic dipole in a magnetic field feels a torque τ given by

$$\vec{\tau} = \vec{\mu} \times \vec{B} \tag{9.8}$$

The angular momentum of a rotating object is

$$\vec{L} = I\vec{\omega} \tag{9.9}$$

where I is the moment of inertia and $\vec{\omega}$ is the angular velocity.

Important Note about Variables Used in This Experiment

Unfortunately, electrical current and moment of inertia are both traditionally represented by the variable I . In this experiment, you will deal with both quantities.

You will have to pay close attention to which I is indicated in each equation.

We can combine these equations for the equation of motion of a magnetic dipole, which also possesses a moment of inertia I , in a magnetic field,

$$\frac{d(I\vec{\omega})}{dt} = \vec{\tau} = \vec{\mu} \times \vec{B} \tag{9.10}$$

Since the vectors point in the same direction, and the angular speed is the time derivative of the angle of rotation, the magnitudes of the vectors obey the following equation

$$I \frac{d^2\theta}{dt^2} + \mu B \sin \theta = 0 \quad 9.11$$

If we assume small oscillations with $\sin \theta \simeq \theta$, this equation is that of a simple harmonic oscillator, with motion

$$\theta(t) = \Theta_0 \cos(\Omega t) \quad 9.12$$

where

$$\Omega = \frac{2\pi}{T} = \sqrt{\frac{\mu B}{I}} \quad 9.13$$

Magnetic Force

The equation expressing the force applied to a magnetic moment $\vec{\mu}$ by a gradient of the magnetic field, $\vec{\nabla} B$ that the moment is contained in is given by.

$$F = (\vec{\mu} \bullet \vec{\nabla}) B \quad (9.14)$$

If the magnetic field and the magnetic moment are both aligned in the z direction, Eq. (9.7) can be simplified to

$$F = \mu \frac{dB}{dz} \quad (9.15)$$

The axial magnetic field gradient for N turns of a single coil can be found by differentiating the expression for the axial magnetic field (9.1) yielding:

$$\frac{dB}{dz} = \frac{d}{dz} \left(\frac{\mu_0 I N R^2}{2(R^2 + z^2)^{3/2}} \right) = -\frac{3}{2} \frac{\mu_0 I N R^2 z}{(R^2 + z^2)^{5/2}} \quad (9.16)$$

This allows us to find the field gradient, since we have the values for N (390), R, and z (D/2). Only the force acting on the dipole is needed to be able to find μ from (9.8).

Since we will be using a spring, it is helpful to review Hooke's Law, which relates a spring's restoration force to the amount that it has been displaced from its equilibrium length:

$$\vec{F} = k\vec{x} \quad (9.17)$$

where F is the force, x is the amount of displacement and k is the spring constant. By making a graph of force versus displacement, the spring constant of the spring can be found from the slope of the graph, which can then be used in calculations to find the force exerted by the spring for any displacement x (assuming that value of x is not too large, the relationship is fairly linear).

Experimental Procedure

Overheating Equipment

After running for a while, the control box may get somewhat warm. This is normal. If, however, your control box gets so hot that it is uncomfortable to the touch, turn the apparatus off and inform your TA. You will need a different control box.

Part I: Finding the Strength of the Magnetic Dipole μ

In the Introduction, it was shown that when a mass that is free to rotate in a gravitational field is placed in a magnetic field with a dipole, the distance the mass is positioned from the point of rotation is proportional to the magnitude of the magnetic field that the dipole is in. This relationship will now be used to find the magnitude of the magnetic dipole moment.

Assemble the Magnetic Torque Apparatus

- The large grey bundle of wires coming from the coils attaches to the 8 slot socket on the back of the control box. There is only one way in which the connector can be plugged in.
- The clear air hose snaps onto the white jack on the back of the control box. A click will be heard when the connector is securely attached.
- Connect the power cord to the control box and plug the apparatus in to a power strip or wall socket. You will also need to flip the switch on the back of the box on. You will know if the unit has power if the red light on the front of the control box lights up.
- Remove the plastic mass from the aluminum rod and record its mass. Record this mass in kilograms and replace the mass.
- Measure the diameter of the ball as well as the length of the ball's handle in meters (See figure 9-2). Record these values on your worksheet.
- Insert the aluminum rod with the plastic mass into the ball. Make sure the end of the rod with the small piece of iron is the end inserted – you should feel the iron

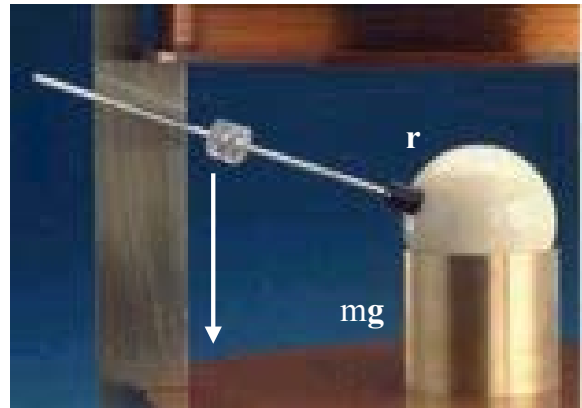


Figure 9-1. The gravitational torque acting on the ball is $\vec{\tau} = \vec{r} \times m\vec{g}$

snap to the magnet in the ball. This plastic mass will be the weight you will be adjusting the position of to balance the magnetic torque.

- Start the air using the switch on the control box. Set the ball and rod onto the air bearing with the rod oriented between the coils as shown in Figure 9-1.
- Set the dial labeled Magnet Current to 2 A. Make sure that the Field Direction is set to UP and the Field Gradient is set to OFF. Set r' to 5 different values between 3 and 10 cm. Then vary the current to get vertical equilibrium for each value of r' . Do not exceed 4 A. If 4 A is not enough current to achieve equilibrium, choose a smaller value of r' . You may need to steady the ball and weight by allowing the rod to rub against one of the vertical supports. Don't be surprised if the system tends to oscillate or drift slightly in the horizontal direction—this is due to the air bearing and the earth's magnetic field acting on the dipole, and you only care about vertical equilibrium.

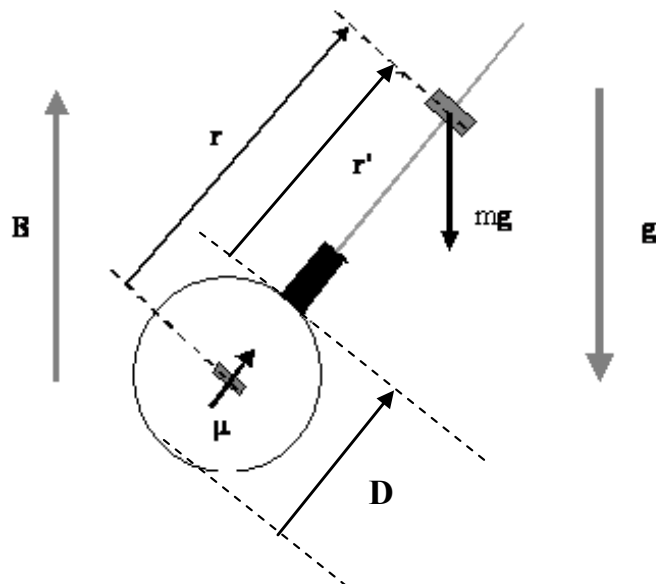


Figure 9-2. Relevant quantities for determining r .

magnitude is calculated from

$$B = 0.00136 \times I \text{ [T]} \quad (9.18)$$

where I is the coil current.

- For each value of r' :
 - Record r' and the current I on your worksheet.
 - From Figure 9-2 you can see that $r = r' + (D/2)$. Determine and record this value on your worksheet.
 - Calculate the magnetic field strength for a given current using equation 9.18. The external magnetic field is created by the Helmholtz coils and its
- Turn off the air and turn the current down to 0.
- Make a graph of distance (r) versus magnetic field strength (B). According to Equation 9.7, the slope of this line is equal to the magnitude of the magnetic moment over weight, μ/mg . Find the slope of your graph, then calculate and record the value of the magnetic moment on your worksheet. Print your graph.

Part II: Simple Harmonic Motion of a Magnetic Dipole in a Magnetic Field

- Set up the magnetic dipole (encased in the opaque plastic sphere) in the Helmholtz coils. Turn on the air flow to minimize friction. Set the current in the Helmholtz coil to 1 A.
- Tilt the dipole to an angle of about 20° and release it. Notice that it undergoes simple harmonic (oscillatory) motion. Measure the period as precisely as possible using a stopwatch to measure the time it takes to swing through many (say 20) oscillations, and derive the period. Make several such measurements and take the average. Note also the error in the measurement, which is roughly the difference between a typical individual measurement and the mean. Record your data on your worksheet.
- Repeat this procedure for coil currents of 2 and 4 amperes. Determine the strength of the magnetic field from Eq. 9.18. Record the data for the magnetic field as well. From Eq. 9.13 we can see that the magnetic moment of the dipole is given by

$$\mu = \frac{4\pi^2 I}{T^2 B} \quad 9.19$$

where the moment of inertia (NOT the current) I is given by

$$I = \frac{2}{5}MR^2 \quad 9.20$$

- Calculate μ and record it on your worksheet. Show your work in a SAMPLE CALCULATION.

Looking at Precession

You know from your studies of physics that an object with angular momentum, subject to a torque, will precess. The magnetic dipole will precess if it is given angular momentum.

- Rotate the dipole so that its dipole axis (marked by the stem on the ball) is at an angle with respect to the magnetic field. Spin the dipole about its dipole axis. Observe its precession. Repeat this observation with different values of the magnetic field. Comment on your observations in Analysis Question 2.

Part III: Finding the Force on a Magnetic Dipole

In this portion of the lab the force exerted on a dipole in a magnetic field will be analyzed.

- Set up the apparatus as shown in Figure 9-3. Note the spring and rod hang from the plastic attachment that sits on top of the tube. The tube in turn rests on the air bearing. Adjust the location of the magnet so that the bottom of the magnet holder is as close as possible to the 2 cm mark on the tube. The air WILL NOT be used for this portion, so make sure the switch is set to OFF.

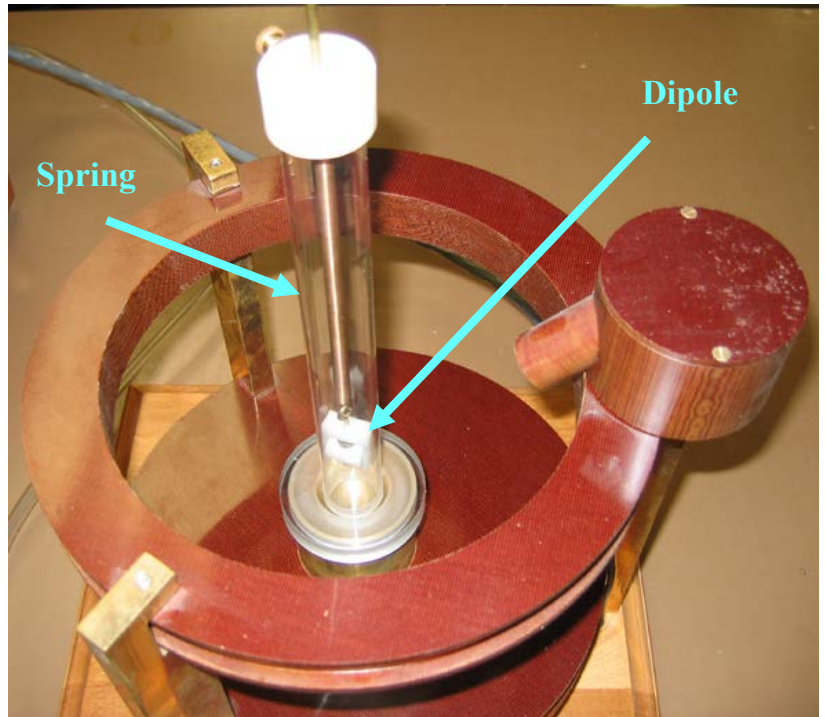


Figure 9-3. Apparatus with dipole hanging from spring.

- Make sure the Field Gradient is switched to OFF and the Field Direction is switched to UP. Bring up the Magnetic Current (and thus the field) to about 4 A. Switch the field direction from UP to DOWN. Continue to alternate the direction of the field and record your observations and an explanation for what you observe on your lab sheet. Be sure to address if a FORCE is being applied to the magnet based on the direction of the field. This can be proven by seeing if the spring is stretched or compressed based on the direction of the field.
- Turn down the current.

Before we can find out the magnitude of the force on the dipole, we need to determine the spring constant of the spring we are using. Here is the procedure:

- Adjust the rod so that the maximum amount is out above the plastic holder. With the spring hanging vertically, note the position of the bottom of the plastic holder for the magnet on the scale.
- Add a ball bearing to the magnet. Note how far the string has been stretched (the difference between the position without a ball bearing and the position with the ball bearing). Record this on your worksheet.
- Repeat for two and three ball bearings. Make sure all your displacements are recorded in the data table.

- Convert the mass of the ball bearings to force using $F = mg$.
- Make a graph of force versus displacement. According to Hooke's Law ($F = -kx$, where k is the spring constant), the slope of this graph is the spring constant. Record this value on your lab sheet.
- We can now use the spring constant to find the magnitude of the force acting on our magnetic dipole for a given magnetic gradient. Make sure all ball bearings are off of the magnet before you begin. Adjust the vertical position of the magnet so that it is equidistant from the coils and note the position of "center" according to the scale on the side of the tube (the position of the "center" should be roughly at the same point as the marked zero on the scale on the side of the tube). Record this on your data sheet.
- Measure the length of rod that is above the plastic cap and record this as L_0 on your data sheet.
- Make sure the magnetic dipole is aligned with the field. With the Field Direction set to UP and the Field Gradient set to OFF, bring the current up to 4 A and make sure the dipole aligns.
- With the current at zero switch the Field Gradient to ON. Bring the current up to 0.5 A. The magnet should displace some amount downward. Adjust the rod so that the magnet is returned to the center of the coils and measure the length of rod that is above the cap. The difference in this length and the original length, L_0 , is the amount the spring has been displaced. Record this value (L) on your data table.
- Repeat this process in 0.5 A increments up to 4 A. Record these values in your data table.
- You may find that you run out of rod for the last current or two. If so, you do not have to take these data points.
- Use your displacement data to find the force applied to the magnet at each displacement length and complete the force column in the data table.
- We can calculate dB/dz using the following relationship:

$$dB/dz = -.01687I \text{ [T]} \quad (9.21)$$

- Make a graph of force versus magnetic field gradient (dB/dz) and find the slope. Print your graph.
- Using Eq. 9.15, determine and record the value of μ .

Analysis Questions

1. Calculate the mean of your values for μ from Parts I and II of this experiment. What value differed most from your mean value? Calculate this discrepancy as a percent difference. Look at the variables which went into calculating μ and discuss which one might cause the greatest error.
2. Give your observations and comments on the precession of the magnetic dipole and its connection to the strength of the magnetic field.